Fusion of functions revisited: When does a head have a second function?

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Abstract

Some grammars give a word two syntactic functions at once. *The Cambridge Grammar of the English Language* treats *some* in *take some* as both the determiner and the head of its noun phrase, and *rich* in *the rich* as both modifier and head, a fusion of functions that Payne et al. (2007) defend. The simpler alternative is that the word is just the head. I argue that a second function needs evidence of form that varies with it rather than with the word, and test for such evidence with the grammar's own diagnostics and new corpus data. For the modifier function there's none. For the determiner function there's one fact, that no adjective comes before a bare determinative, but the comparison is close, and on balance it doesn't justify the extra function. Fused relatives such as *whoever told you that* are different: they relativize subjects, they have no infinitival forms, and in usage the case of *whoever* answers to its function inside the clause. There fusion is favoured, though not forced. Keeping a category, heading a phrase, and taking a reading from context are things an ordinary head can do, not evidence of a second function.

Keywords: determinatives; fused heads; free relatives; headedness; noun phrase; quantifiers; English grammar

AI use. GPT-6 (Astra), GPT-5.6 (Sol), Claude Fable 5.1, Opus 5 and 5.5, and Haiku 4.5, GLM-5.3-Flash, and Muse Wonder 1.0 were used through their vendors' apps and APIs, April to October 2026, to help draft and revise prose, check sources, and classify corpus lines (Section 2), and to simulate peer review. I checked their output and am responsible for the paper.

1 Introduction

When one expression seems to have two functions at once, what should persuade us that it does? In *some apples*, *some* determines the noun phrase, and the noun *apples* is its head. In *I'll take* [*some*], *some* is the whole noun phrase, and there's no noun to be its head. One influential analysis gives *some* both functions there: it's the determiner of its noun phrase and the phrase's head at once. This paper asks what evidence could justify that doubling.

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The doubling is called FUSION OF FUNCTIONS. Its most explicit statements are in *The Cambridge Grammar of the English Language* (*CGEL*; Huddleston and Pullum, 2002) and in Payne, Huddleston, and Pullum (2007; henceforth PHP). In *take some*, *CGEL* analyses *some* as a fused determiner–head. In *the lucky few* and *the rich*, *few* and *rich* are fused modifier–heads, at once modifier and head (pp. 410–421). PHP make the device explicit, extend it to *once*, *twice*, and *thrice*, and say what it costs.

It departs from canonical structure twice: a constituent has two functions, and a phrase can have a head of another category, as when a noun phrase is headed by a determinative or an adjective rather than a noun (PHP, pp. 570–571). The question is when a grammar should accept the first.

Many syntacticians would sidestep it. The determinative could head the whole noun phrase, as on the DP hypothesis (see Miller & Pullum, 2022), or a silent noun could follow *some* (Kester, 1996; Lobeck, 1995). Each route gives up one of three commitments that *CGEL* argues for and this paper keeps, since they're the premises that led to fusion (Section 2). They don't decide between fusion and the alternative introduced below, so the choice falls to evidence of form. The result is conditional on those commitments, but the criterion can be applied to a second function in any analysis.

The alternative I call ORDINARY HEAD. The phrase that *CGEL* fuses has one function, head of the NOMINAL, as a common noun does in *take* [*apples*]. The nominal is the unit between noun and noun phrase, such as *old man* in *the old man*. Since the two analyses describe the same strings, choosing between them takes a criterion, and I put the burden on the second function. A property of form counts as evidence for it if it varies with that function and not with the word alone, and the evidence is weighed against the extra structure (Section 2).

The test finds no such property where a word that's elsewhere a modifier heads a nominal, as in *the lucky few*, *the rich*, and *nylon* in *I prefer cotton shirts to* [*nylon*] (Sections 2 and 4), and *once*, *twice*, and *thrice* add nothing further (Section 5).

For a determinative standing alone, the test finds one, the absence of an adjective before it (**Lucky few escaped*).[1] Fusion derives that, where ordinary Head states a condition of a kind *CGEL* already states for pronouns (Sections 3 and 4). Section 6 weighs the advantage against the fewer functions ordinary Head assigns and the constituents with two parents it avoids, and I take the balance to favour ordinary Head, more clearly for the modifier function than for the determiner function, though not to force it.

Fused relatives are another matter (Section 7). In [*Whoever told you that*] *can't have read the report*, *whoever* is head of the subject noun phrase and also the subject of *told you that* (*CGEL*, p. 1076). On the alternative, the head analysis, the relative word simply heads the noun phrase and the clause modifies it (Bresnan & Grimshaw, 1978). But the second function shows in form, twice categorically and once as a preference. Fused relatives relativize their subject, which bare relatives (those with no relative word) can't do in standard English. They're never infinitival, though bare relatives can be. And in usage the form of *whoever* answers to the function inside the clause as well as to the function of the whole phrase. Fusion gets all three from the rules for relative clauses with a relative phrase, where the head analysis needs statements that apply to no other head. So fusion is favoured there, though not forced, and it's the one construction where this paper keeps it.

[1] As in *CGEL*, * marks an unacceptable form, ?a questionable one, !a non-standard one, and %one found only in some varieties.

2 Two analyses and a criterion

The framework keeps categories apart from functions. A category is a set of words or phrases, such as noun, determinative, or adjective. A function is the relation a constituent bears within its phrase, such as head, determiner, or modifier. The two vary independently: *nylon* is a noun functioning as head in *made of* [*nylon*] and as modifier in *a* [*nylon*] *shirt* (*CGEL*, p. 421), and an NP functions as determiner in [*Kim's*] *book*.

So DETERMINATIVE names a category (*the*, *this*, *some*, *every*, *many*), and DETERMINER names a function. Figure 1a gives *CGEL*'s structure for *some apples*. The determinative heads a determinative phrase (DP) that functions as determiner (Det) of the NP.[2] The noun heads the NOMINAL (Nom), the unit between noun and NP, and the nominal is the NP's head. *CGEL* recognizes the unit because *old man* in *the old man* is neither a noun nor an NP, since it can't function as subject or object (p. 329). I write "nominal" in prose and Nom in trees and quotations.

The NP's ULTIMATE HEAD is the word reached by following head relations down from the NP: *apples* in Figure 1a, *some* in b and c.

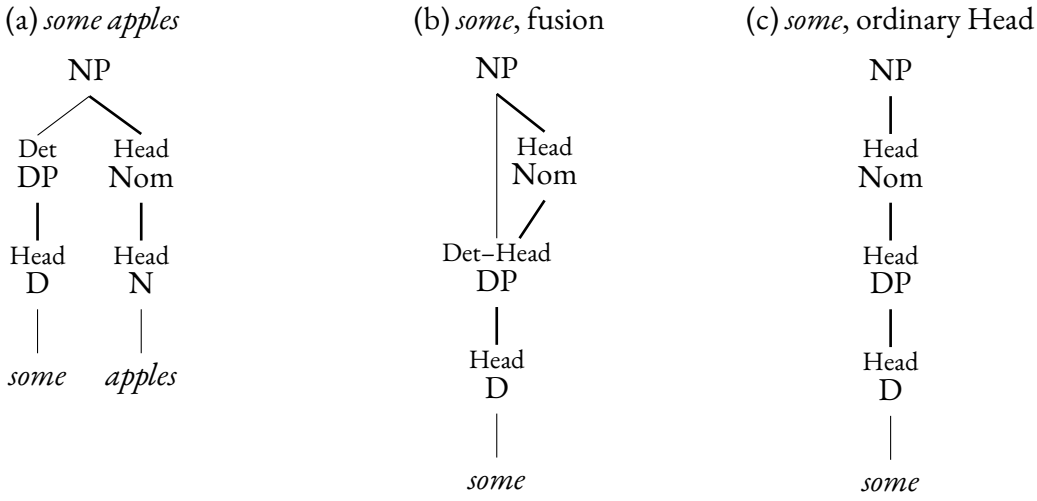


Figure 1: *Some apples* (a), and *some* standing alone under fusion (b) and ordinary Head (c). Function labels sit above category labels, and thick lines join heads to their parents. In (b) the DP has two parents, with a line from the NP for its determiner function and one from the nominal for its head function. In (c) it's the head of the nominal and nothing else. Each tree shows the DP even where it contains only D, so that the same constituent appears in all three.

Without a noun, as in *I'll take* [*some*], the nominal still needs a head. *CGEL*'s answer is FUSION OF FUNCTIONS: the DP functions as determiner and as head at once, a fused determiner–head (Det–Head; Figure 1b) (pp. 410–421). After a determiner, as in *the lucky few*, the word is a fused modifier–head (Mod–Head). PHP make the device explicit and list where English uses it: "The subtypes of dependent involved are determiner, modifier and, in fused relative clauses, subject or prenucleus"

[2] DP here is *CGEL*'s determinative phrase, a phrase whose head is a determinative, not the DP of the DP hypothesis, on which the determinative heads the whole noun phrase. Miller and Pullum (2022) draw the same distinction. PHP (p. 566 n. 2), like *CGEL*, take the noun as head of the NP.

(p. 571; the prenucleus is the position of a fronted relative phrase). Sections 3, 4 and 7 take them in that order, and Section 5 takes PHP's own case, *once*, *twice*, and *thrice*.

Fusion, as PHP define it, departs from canonical structure in two ways. First, one constituent has two functions, so that it "will in effect have two mothers", which they call "the major departure from canonical principles of structure" (p. 570). The structure is then a directed acyclic graph rather than a tree (Pullum & Rogers, 2008). In Det–Head the shared DP has two parents, the NP and the nominal; in Mod–Head both functions belong to one nominal (pp. 582, 588). Second, an NP can have an ultimate head of another category: the fused node is "a point in the normal path of headedness at which there can be a change of category" (p. 569).

Ordinary Head keeps the second departure and drops the first. Whatever phrase *CGEL* fuses functions as head of the nominal and nothing else: a DP for *some* (Figure 1c) and for *few* in *the lucky few*, an AdjP in *the rich*. For adjectives this is close to the traditional analysis, though there the adjective heads the noun phrase itself: "Adjectives can function as heads of noun phrases" (Quirk et al., 1985, p. 421). A determiner before it, as *the* in *the rich*, is the NP's determiner as usual, and the nominal keeps its other dependents: in *few of them* the *of* -phrase is a complement in the nominal headed by *few*, as in *CGEL*'s own partitive tree (pp. 411–412). What fusion states as a licence for a fused function, ordinary Head states as a permission: a word may head a nominal, with or without a determiner before it. *Some* may head one with no determiner, and *rich* one after *the*.

CGEL calls a head with no second function a pure head (p. 410), and it considers the analysis itself, as "change of function without change of category", with *many* a determiner in [*Many*] *people would agree* and a head in [*Many*] *would agree* (p. 421).

The change of category needs no new kind of structure. PHP say it "will be independently required in the case of hybrid constructions" (p. 571), and their tree for *there's* [*no stopping him*] has a VP as head of an NP whose determiner is *no*: "an unexpected category appears as head … However, there is no fusion of functions in this case" (p. 600). The tree shares the two properties that matter here, an unexpected category as head and no constituent with two functions, though the VP heads the NP directly, with no nominal between. PHP find hybrids "only marginally in English" (p. 566), which might seem to count against ordinary Head. But ordinary Head changes category only where fusion does, and at the same node, so it uses no departure that fusion doesn't also use, and uses it no more widely.

Both analyses keep three commitments that *CGEL* argues for, and the comparison holds them fixed. First, in *some apples* the head is *apples*, not *some* (Figure 1a). The noun is the NP's ultimate head, and the determinative heads a DP that functions only as determiner. Second, the determinative keeps its category when it stands alone: it takes the dependents it takes before a noun (*not much cheese*, *not much of the cheese*), which "argues against any part-of-speech distinction between the two uses" (*CGEL*, p. 420). Third, there's no silent head noun. Analyses that give up one of these avoid fusion by other means: a determiner phrase headed by the determinative, or a silent noun.[3]

This paper doesn't take those analyses on. Its result is conditional: given *CGEL*'s commitments, the balance favours fusion only in fused relatives. Holding *CGEL*'s categories fixed isolates the ques-

[3] On the DP hypothesis the determinative heads the phrase (see Miller & Pullum, 2022). A silent noun is recovered from an antecedent on ellipsis analyses (Lobeck, 1995) and needs none on empty-noun analyses (Aarts, 2024; Kayne, 2007; Kester, 1996; Saab, 2019). Wu (2021) puts an empty noun after *the rich* and after a determinative standing alone (pp. 19–20, 118, 173). *CGEL* gives its grounds against a silent head on pp. 420–421.

tion of functions; it doesn't endorse them. Indeed, Reynolds (2026) argues that determinatives are a kind of noun, beside common nouns, proper nouns, and pronouns. On that classification a determinative heading a nominal is a noun heading it, so the second departure disappears for determinatives on both analyses, leaving ordinary Head with no departure for them and fusion with the first.

Since both analyses describe the same strings, the comparison needs a criterion. Interpretation separates analyses that predict different readings, as it separates both of these from a silent *people* (Section 4). But fusion and ordinary Head predict the same readings, from the same sources.

CGEL classifies fused heads twice, by function and by interpretation (p. 333). In its simple type a head recoverable from an antecedent could be added (*Four boys played croquet and* [*two*] *played tennis*, 'two boys'), in the partitive type an *of* phrase, stated or understood, gives the set (*He made ten mistakes but* [*most*] *were fairly trivial*), and in the special type the reading is fixed for the word (*the rich*, 'rich people'). The classifications cross: *CGEL*'s examples put Det–Head in all three types and Mod–Head in the partitive and the special.

CGEL also gives these readings to a head with no second function. *Other* heading a nominal has a plural, *others*, so *CGEL* takes it to be a noun, with the structure "simply determiner + head" and "no syntactic fusion of two functions" (pp. 1517–1518): *the others*, 'other boxes'; *the other*, 'the other one of the plates'; *others*, 'other people'.

So the reading comes from an antecedent or the situation, from a partitive, or from a convention that both analyses list word by word (*CGEL* lists *few*, *many*, *all*, and *some* as taking 'people', p. 414). The second function supplies none of these. Each analysis still needs a route to the reading, and interpretation would distinguish them only where one needed a process the other doesn't. What's left is form, which is what *CGEL* rests the determiner function on when it argues from the genitive (p. 420). PHP call a constituent with two functions a departure from canonical structure, so I take the burden to lie with the second function.

Where the same category and head relation account for an expression, a property of form counts as evidence for a second function if it varies with that function and not with the word alone. By form I mean what can be observed of an expression: the shapes of its words (a case such as *whom*, or a variant such as *none*, which occurs only without a noun after it), their order, and which dependents and determiners occur with each. Interpretation alone isn't form.[4]

The criterion answers *CGEL*'s objection to ordinary Head. *Nylon* in *I prefer cotton shirts to* [*nylon*] is one of *CGEL*'s fused modifier–heads, of the type denoting composition (p. 417). *CGEL* rejects analysing it as a noun functioning simply as head, since that "would not distinguish this construction from *I prefer cotton to nylon*", where *nylon* isn't "understood as having modifier function" (p. 421). The difference is real. But the word is the same in both sentences, and so is its form: bare *nylon* as the object of *to*. What differs is the antecedent. In the first sentence *cotton shirts* supplies the shirts that *nylon* is understood to describe, and *nylon* describes a shirt in its own meaning, as in *This shirt is nylon*. Fusion needs the antecedent too, since the modifier function doesn't supply the shirts.

The one formal property *CGEL* ties to these modifier–heads is their determiners. The NPs of *nylon*'s type "are either bare or have a definite determiner", with "significant loss of acceptability

[4] *CGEL* reasons this way when it declines extra structure. It doesn't take a content clause after a verb to be an object, since the clause isn't "sufficiently like an NP" (p. 1018): it needn't follow the verb directly (*He denied categorically that he had spoken to her*, but **He opened slowly the door*), and some verbs take it but no NP (*marvel*, *vouch*). A bare determinative patterns with the NP (**He took slowly some*; *I see some*, *with some*; Section 3), so the criterion doesn't favour less structure as such.

when the determiner is indefinite" (p. 417), and nouns of other types are excluded, as in **Bill likes the linguistics lecturer, but I prefer* [*the sociology*] (p. 415). The examples in (1) cross both lines.

(1) a. *You take the cat shirt and I'll take* [*the dog*].
b. **You take the cat shirt and I'll take* [*dog*].
c. *You take the cotton shirt and I'll take* [*nylon*].
d. *a misdemeanor sentence rather than* [*a felony*] (*Arizona Daily Star*, 2019, COCA)

Dog and *felony* belong to no type *CGEL* lists, and they work. *The dog* needs one step more than *nylon*: the relation between dog and shirt (bearing a picture of one) has to be found, as it does for *cat* in *the cat shirt*, and fusion leaves the same relation to be found.

Bare *dog* is out and *a felony* is fine, as for a singular count noun generally. Non-count *nylon* stands bare for a single item and resists *a* (?*I bought a cotton shirt rather than* [*a nylon*], p. 417). So the determiners follow the noun's own countability. Both analyses make the noun the head, so both predict that. Neither kind of silent head predicts that without a further provision: every antecedent in (1) is a count noun, so a head that took its features would reject bare *nylon*, and a head with none of its own wouldn't exclude bare *dog*. What the facts don't show is anything that varies with the modifier function, and the restriction *CGEL* states for it doesn't hold.[5] On this criterion, nothing here counts as evidence for the modifier function.[6]

The criterion says what counts as evidence for a second function, not how much is enough. A fact counts if it varies with the function and the word alone doesn't explain it. Whether it outweighs the extra structure is a separate question, and a fact that something else could also explain weighs less. So the later sections compare the analyses construction by construction: what each predicts, where the data go against a prediction, what each has to state instead, and what the extra structure buys. Where the data can't separate the analyses, I say so, since finding no evidence for a second function isn't finding evidence against it. *CGEL* itself names a form that would count: a fused determiner–head should resist the genitive, because "for full acceptability its case should match both functions" (p. 420).

A note on data. Examples are *CGEL*'s or PHP's where cited, attested where a source is given, and otherwise constructed. Judgments not attributed to a source are mine, and claims about what's excluded are for standard English. Corpus lines come from COCA, the Corpus of News on the Web (NOW; Davies, 2016–present), and the iWeb corpus (Davies, 2018), searched by hand at english-corpora.org in September and October 2026; where a display was capped, counts are of the lines shown. Some sets of lines were classified by a large language model (Claude Opus 5.5), and each line's label is kept so that the classification can be checked. The genitive lines in Section 3 were labelled twice, independently, with the same result. I checked the labels behind Sections 3 and 7 by hand (for

[5] Dependents give the same result. Matthews (2014, pp. 28–29) notes that material nouns used as modifiers take *partly* and *largely* (*a partly cotton shirt*) and suggests they're adjectives there. But uses with no modifier function take them too (*I prefer cotton to* [*partly nylon*]; *This lava is* [*partly magma*], conserve-energy-future.com), so *partly* isn't tied to the modifier function.

[6] *CGEL*'s other example, *Mary earns* [*double*] (p. 421), fuses a noun with a predeterminer modifier (p. 418), and *double* gets the treatment *nylon* gets. Its category fits. Before a definite determiner, *CGEL*'s predeterminers are nouns and NPs (*half the cake*, *double the number of applications*) or the adverb *twice* (p. 434), and my searches of the Corpus of Contemporary American English (COCA; Davies, 2008–present) found no adjective phrase there, apart from one marginal *400-fold the amount* (a blog).

the earlier searches, a sample of 105) and changed five, none affecting a figure the paper reports. The record behind each corpus claim, with its query, date, counts, and labels (but not the corpus lines), is listed in an evidence index.

3 Determinatives standing alone: Det–Head

A determinative can make up a whole noun phrase, or head one that also contains a complement or modifier, and so can an independent genitive such as *mine* (e):

(2) a. *I'll take* [*some*].
b. [*Few*] *would dare underestimate Aaron Finch's men.*
c. *some of the wine*, *many of them*, *whichever of them*
d. *hardly anyone present*, *something reliable*
e. [*Mine*] *are ready.*

For *CGEL* each of these has a fused Det–Head: one phrase functions at once as determiner of the NP and as head of its nominal (pp. 410–423). On ordinary Head it's a pure head, with no second function (Section 2). The question here is whether anything in the form of these expressions shows the determiner function.

Some things aren't at issue. Both analyses take these to be complete NPs with no silent noun. They have the functions of NPs (*Some left*, *I see some*, *with some*), there's no noun to restore (**many ones of us*; pp. 420–421), and often there's none in the context either.[7] Nor is the lack of a further determiner evidence for the determiner function. *CGEL* gives *She left*, *Kim left*, and *People left* NPs with no determiner, and ordinary Head gives *Some left* the same structure. Nor is definiteness: bare *this* and bare *some* differ in it as they do before a noun, so the word carries it on either analysis.

Where the context names no set, as in (2b), a domain the context supplies restricts it, as it restricts any quantified expression, even one with a noun; for *few*, *many*, *all*, and *some* a convention also gives 'people' (Section 2). Von Fintel (1994, p. 43) represents the domain as an index on the quantifying word, not a constituent, so it isn't the silent noun set aside in Section 2, and it says nothing about which determinatives can stand alone; both analyses list them (Table 1).

Partitives add nothing here. In *some of the wine*, the *of* -phrase is a complement inside the nominal (*CGEL*, pp. 411–412), so *some* is the ultimate head on either analysis. The claim about bare uses extends to them unchanged, including *whichever of them* in the fused relative of Section 7. So do relative clauses,[8] and complements licensed by a noun in the antecedent, as in *We have a proposal to close the petition and* [*another to keep it open*] (Scottish Parliament, Public Petitions Committee, *Official Report*, 16 December 2008).[9]

[7] Of 94 NOW lines for *few would*, 82 have no noun phrase or named set in the displayed context that could supply the set, as in (2b) (*Daily Times*, Pakistan).

[8] Wu (2021, pp. 184–185) finds a dilemma for fusion in *For* [*some who made the journey into Wales*], … (BNC): fusion makes *some*, a determiner, the antecedent of the relative clause, against the principle that determiners don't count as antecedents, and he adds an empty noun to resolve it. On ordinary Head, *some* heads the nominal and the relative clause modifies that nominal, with nothing added.

[9] The infinitival is a complement of the kind *proposal* licenses (*CGEL*, p. 1259), so on either analysis its licence, like its reading, comes from the antecedent. *CGEL* has a PP complement in the same position in *Helen's of the Rachmaninov* (p. 421). The antecedent licenses the complement but not the head: **Their names weren't on the list of the dead, nor on the of the missing* is out, with *the* read as 'the list', because *the* doesn't stand alone (p. 1508).

CGEL and PHP point to three places where the determiner function might show: the correspondence with the determiner use; the forms of *none*, *mine*, and the compounds; and case. None of the three shows it, but one fact of distribution, which comes up with the compounds, favours fusion.

First, the correspondence. *CGEL* keeps the determiner function because the word "can occur in the dependent position in NPs where it is not fused with the head" (p. 410). It keeps the category because the word takes the same dependents in both uses (*not much cheese*, *not much of the cheese*; pp. 420–421). Ordinary Head keeps the category, and with it both facts: *much* is the same determinative in both uses, and it takes the dependents a determinative takes.

What the correspondence adds beyond the category is a list of the determinatives that can stand alone. *Some*, *each*, and *either* can, and *every* normally can't (**Every of the apples was rotten*, p. 413), though NOW has two published lines with it standing alone, whose status I leave open (*Every has a story*, *Patch*; *Every has different blocking mechanics*, *Guardian*); *no* does so only in the form *none*. PHP (p. 581) state the list as a "lexical distinction between determinatives that allow [fusion of functions] and those that do not". Ordinary Head states the same list as a permission to head a nominal with no determiner. Neither analysis gets it free.

Second, form. PHP find formal differences between fused and non-fused functions here, as in fused relatives: *no* "has a specialized morphological variant *none* which is restricted to the fused function" (p. 581). A difference of form doesn't say which function it marks, though, and the genitive pronouns supply the test. They alternate the same way (*my*, *mine*), and *CGEL* states the conditioning for both pairs at once: "the dependent forms are used in (pure) determiner function, the independent ones in head function" (p. 410).

Mine occurs with a determiner function and without one. The independent genitive pronouns "occur in fused heads and also pure heads" (p. 410), pure heads being heads with no second function, and predicative *mine* in *Everything that is* [*mine*] *is yours* has "no basis for a fused-head analysis" (p. 469). Its form is the same, so the form doesn't mark the determiner function. *None* has no pure-head use to test (p. 410), so its form is assigned to the same conditioning on the model of *mine*, not shown independently. Ordinary Head then states both alternations as *CGEL* does, by function: Det against Head.

Compounds raise the question of form again. *CGEL* analyses *anyone*, *something*, and the rest as fused determiner–heads (p. 423). PHP take them to be determinatives because "they take exactly the same pre-head modifiers as the determinative bases they contain" (p. 581): *not everything* beside *not every house*, *hardly anyone present* beside *hardly any writer present*. That fixes the category, which ordinary Head keeps.

On ordinary Head, *not* and *hardly* are modifiers in the DP that heads the nominal, as they are in the DP functioning as determiner in *hardly any writer present* (*CGEL*, p. 431), so the compound's base still decides which ones it takes. PHP also take the compounds' morphology, a determinative base with a nominal base, to mirror fusion (pp. 565, 587–588). The two bases can as well reflect the compound's meaning, a quantifier with a sort ('person', 'thing'), and both analyses give it that meaning. Simple *some* and *few* have no nominal base, and for PHP they're fused just the same.

The compounds' modifiers divide by side: adverbs before, adjectives after (*hardly anyone present*, *something reliable*). The adjective is a RESTRICTOR, a single post-head dependent that doesn't recur and can be a simple adjective, of the kind that would normally precede a noun (p. 423; PHP, p. 583). PHP count fusion's account of the division "a particular strength" (p. 583). On ordinary Head the

compound's DP heads the nominal and the restrictor follows it in the nominal, so the two analyses group the phrase alike, as [[*hardly anyone*] *present*].

Nor does fusion account for the restrictor's position by itself. *CGEL* puts the adjective after the head "because determiner and head are fused", which rules out internal pre-head dependents (p. 423). But the restrictor stays where nothing is fused. *CGEL* treats *something* in *a little something* as a noun formed by conversion (p. 423 n. 43), and a single adjective follows it (*a little something extra*), as one does in *a pre-emptive something fabulous* (CBS). With a common noun as head, a postpositive adjective occurs only "under restricted conditions" (p. 529).[10] So fusion isn't necessary for the position.

Fusion isn't sufficient for the position either. Simple determinatives standing alone are fused too, and *all*, *much*, and *little* take a restrictor (*little new*), as they share *else* with the compounds (p. 423). But after *few*, *some*, or *none* a single adjective follows only if it's comparative or one, like *present*, that follows nouns too (*I've heard worse but none weirder*, film script; *None present objected*; **There are few good*). A defender of fusion can let the converted noun inherit the restrictor and list the words that take it, and then the post-head side gives fusion no advantage beyond the pre-head one.

The pre-head side does favour fusion, and it's the only fact in the distribution of these words that does. Bare determinatives take no pre-head adjective (**nice anyone*, **Lucky few escaped*), apart from the compounds and partitive *all* and *both* in a few exclamations such as *Poor everybody!* and *Poor all of us!*, and compounds converted to nouns (*instant everything*; Section 4). With *few* as both determiner and head, an adjective modifying the nominal would come before the NP's determiner, where nothing in the nominal can go, so fusion derives the gap from PHP's condition that the fused functions are adjacent (p. 571); ordinary Head has to state it (Section 4), as an analysis with a silent head does (PHP, p. 584). The derivation needs the bare word to be a fused determiner–head, not a modifier–head like *CGEL*'s *blue* in *I like blue* (p. 417), and for *many*, as in *many of us*, *CGEL* gives no argument that picks determiner over modifier. What rules out a bare modifier–head is that the word's use for a group never occurs bare, a fact ordinary Head needs too (Section 4).

The adjective is fine once a determiner precedes (*the lucky few*). It's also fine in *a lucky few*, which *CGEL*'s analyses of *a few* allow two ways (pp. 353, 392). Section 4 shows that on either of *CGEL*'s treatments of *a few*, *lucky* doesn't modify a bare determinative, so the gap stands.

Third, case, where *CGEL* itself says the determiner function would show. In *many people's expenses*, the genitive *'s* attaches to *people*, the head of *many people's*, and not to its determiner, *many*. On the fused analysis a bare *many* is both head and determiner, so **many's expenses* would put the genitive on *many*, which never carries it as a determiner, though determiners can themselves be genitive (*my*, *Kim's*). *CGEL* reports that very gap, calling *all's*, *many's*, *few's*, and *some's* "completely excluded". Its explanation is fusion: "the form is simultaneously determiner and head, and for full acceptability its case should match both functions" (*CGEL*, p. 420).

CGEL offers this against analysing these words as pronouns, an analysis that gives no reason for the gap (p. 420). Ordinary Head gives no reason either, so the argument bears on it too, and it's the most direct of the three.

[10] In COCA, 43 of 46 relevant lines of an article, an adjective, *somebody* or *something*, and a second adjective are *a little something* (*extra* in most). The evidence beyond the formula is thin, and none has a personal compound.

For *all*, *many*, *few*, and *some*, the searches found the gap nearly complete,[11] but it doesn't hold across the fused determiner–heads. *Either's*, *neither's*, *each's*, and *another's* occur in edited prose, and I find them acceptable: *increase* [*either's*] *share of its market* (*TIME*), [*Neither's*] *numbers are all that encouraging* (*Washington Post*), *the highlight of* [*each's*] *day* (*Atlantic*). *CGEL* calls *each's* "occasionally attested but of somewhat questionable status" (p. 420). COCA's twelve genitives of *each* bear out "occasionally attested", but five are in edited prose, so they aren't confined to informal writing.

CGEL's own examples of genitive marking on the head include *everyone's* and *somebody's* (p. 479), compounds it analyses as fused determiner–heads (p. 423). A defender of fusion could put their genitive down to the nominal base, *-one* or *-body*.

So the genitive divides these words, and the argument from case faces a dilemma. If *many's* is out because its case would have to match two functions (p. 420), then *either's*, *neither's*, *each's*, and *another's* should be out too, since *CGEL* gives those words the same two functions. If something else is needed as well, the gap gives fusion no demonstrated advantage. Either way, the shared functions don't predict where the division falls, though function may still play a part. As stated, neither analysis predicts it: fusion has to exempt *either*, *neither*, *each*, *another*, and the compounds unless their nominal base carries the genitive, and ordinary Head has to list the words that don't take it, as it lists the reflexive pronouns, which have no genitive either (p. 426).

The division may follow meaning: the words that take the genitive, the compounds included, are singular, and those that don't are words of quantity (*all*, *many*, *few*, *some*), and if that decides, neither analysis needs a list. A comparison that could show a function would keep the word and change only that function, and these words offer none, since *CGEL* gives them no use as a pure head.

Section 7 finds that kind of evidence in fused relatives, where the form of *whomever* follows the function inside the relative clause.

Apart from the pre-head adjective, then, both analyses need further statements of the same facts, and nothing else in the form of these expressions shows a determiner function.[12] That one fact is a real advantage for fusion, and Section 6 weighs it against what ordinary Head gains.

4 Modifiers as heads: Mod–Head

In Mod–Head fusion the head of a nominal is also its modifier. *CGEL* uses it for words that function elsewhere as modifiers in a nominal: determinatives and adjectives, and also nouns such as *nylon* (p. 417), taken up in Section 2:[13]

(3) a. *This copy is defective but* [*the other two*] *are fine.*
b. *I didn't like it, but I respect the views of* [*the many who did*].
c. *Are you one of* [*the lucky few who found powdered infant formula*]*?*

[11] COCA searches for each word followed by *'s* find genitives on *either*, *neither*, and *each* standing alone in 35 lines, 13 of them in edited prose, against four for *many* and *some*, all on the web (two seem fine to me: *to* [*many's*] *suprise* [sic], *out of* [*so many's*] *reach*), none for *few*, and none for *all* in a capped display; the counts aren't adjusted for how often each word stands alone. *Another* followed by *'s* and a noun, not preceded by *one*, has 1,331 tokens, 798 in edited prose; *CGEL* mentions the genitive only for reciprocal *one another's* (pp. 428, 1502).

[12] Licensing *ever* doesn't show it either. *CGEL* separates determiner from modifier *few* by clause negativity (p. 394), but the adjective *scant*, a modifier for *CGEL* (p. 556), licenses *ever* in its clause when no determiner precedes it, as determiner *few* does (*scant regard was ever given to the people who lived there*, *Northern Daily Leader*). Whether bare *scant* passes *CGEL*'s other tests for a negative clause is untested.

[13] Examples (3a, b, d–f) are *CGEL*'s (pp. 416–419); (3c) is from NOW (*National Observer*).

d. [*The rich*] *cannot enter the kingdom of Heaven.*
e. *How will it affect* [*the very poor*]*?*
f. *There were two pieces left and Kim chose* [*the larger*].

For *CGEL*, *two*, *many*, *few*, *rich*, *poor*, and *larger* each modify the nominal and head it at once (pp. 415–419). PHP (pp. 579–580) draw the adjectival case for *the very hard of hearing*: the AdjP is "head of Nom" and "modifier of that head", and *the* is the NP's determiner as usual.[14] Unlike Det–Head, both functions belong to the same nominal (p. 588; Figure 2). Their main Mod–Head case, *this once*, is taken up in Section 5. On ordinary Head the AdjP or DP is head of the nominal and has no other function.

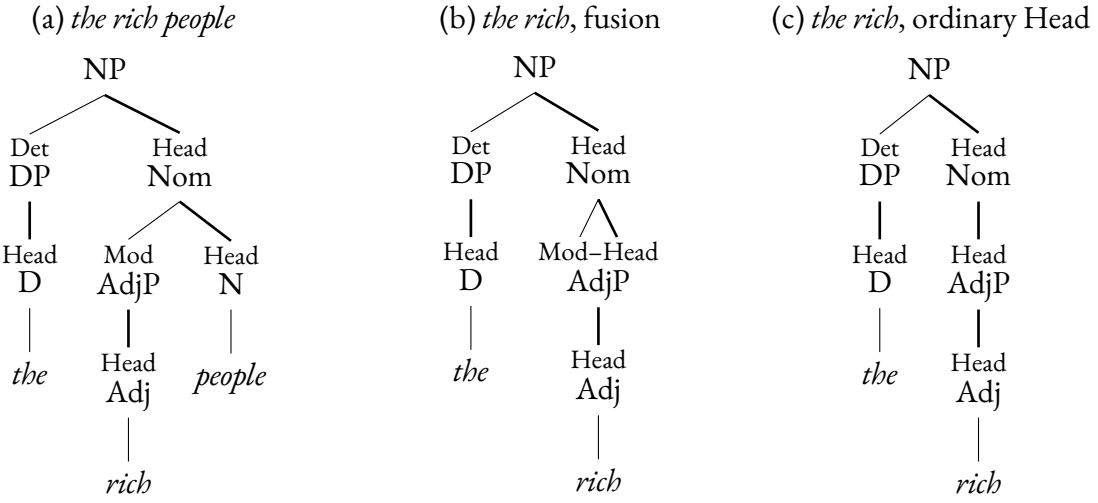


Figure 2: *The rich people* (a), and *the rich* under fusion (b) and ordinary Head (c). In (b) the AdjP has one parent, the nominal, with a line from it for each of its two functions; in Figure 1b the fused DP has two parents. In (c) the AdjP is the head of the nominal and nothing else.

So the two analyses agree that the adjective or determinative heads the nominal, and both keep the word's category. *Rich* and *poor* take adverbs (*the extremely rich*, *the very poor*), which *CGEL* calls "clear evidence that they retain their category status as adjectives", and *the rich* is plural with no plural ending on either analysis (p. 418). Both give the head the dependents it takes elsewhere: a partitive in *the biggest of them* (p. 416), a post-head phrase in *the pure in heart* (p. 418). *Few* takes a partitive with a determiner and without one (*few of them*, *the few of us who stayed*), so its dependents don't distinguish its Det–Head use from its Mod–Head use. And neither has a silent noun, and adding one changes the reading: *the people very hard of hearing* can't be read generically, as *the very hard of hearing* is (PHP, p. 580).

CGEL says the same of *the rich people* (p. 420). Glass (2019) adds readings that a silent *people* would miss: ones not about people (*The weak and mutated die, the healthy survive*, p. 10), and ab-

[14] PHP's tree labels the fused function Head–Mod, since the counterpart modifier follows the head (*people very hard of hearing*); their tree for *this once* has Mod–Head (p. 588). *CGEL* allows fusion with a post-head modifier as well (p. 418). I write Mod–Head for both.

stract ones, singular like *CGEL*'s *the immoral* (p. 418), as in *The crazy … is easily detected* (p. 7). She derives both from the adjective's own meaning (pp. 12–13).

What's at issue is only the second function, the modifier. Evidence for it would be a form that marks the modifier function, a fact of order that only a modifier relation predicts, or determiners or a case that vary with the function. The formal differences PHP cite are in Det–Head and in fused relatives (*none*, p. 581; the *-ever* forms, p. 576), and they cite none for Mod–Head. Their fact of order is adjacency: with modifier and head one constituent, nothing can come between them, so **the in the room very hard of hearing* is out (p. 580 n. 17). That argument is aimed at a silent head. On ordinary Head the PP would be a pre-head dependent in the nominal, which a PP never is (**the in the room people*).

For determinatives, *CGEL* links the fused use to the dependent one. Determinatives "which can function as internal modifier as well as determiner can in general occur as fused modifier-head as well as determiner-head" (*CGEL*, p. 415): *the few* beside *the few mistakes*, where *few* is modifier (p. 392). PHP (p. 588) state the link without the hedge: *some*, *every*, *any*, and *no* "can function solely as determiners and therefore never permit a preceding determiner".

Yet *some* is attested after *the*, if marginally: [*the some*] *will become the many*; *one of* [*the some who hope for a return to the good old days*].[15] With *the* as determiner, a fused *some* can only be a modifier–head, and that's the Mod–Head PHP's "never" excludes. *CGEL* hedges the licence ("in general") and itself records words that head without modifying and words that modify without heading. *Both*, which "cannot occur in internal modifier function", heads a nominal after *the* in %*the both of us* (p. 377). *Every* and *various* function as modifiers after a determiner (*her every move*, *the various options*) but don't head a nominal there: "*Every* is again excluded, and so too this time is *various*" (p. 416).

So the link between modifying after a determiner and heading a nominal after one is a default, not a condition, and the exceptions run both ways. Fusion builds one direction in if it licenses a fused modifier–head through the word's modifier use, which is what keeps its licences general (Section 6): a word with no modifier use then can't head a nominal after a determiner. But *CGEL*'s own *the both of us* is an exception in just that direction, and so, marginally, is *the some*. Both analyses list the determinatives that can head a nominal after a determiner, ordinary Head as permissions, as for those that stand alone (Section 3).

Which determiners NPs like *the lucky few* take depends on the head word, not on a modifier function. With an adjective before *few* and no noun after it, the determiner in the COCA lines searched is *the*, a demonstrative (*those chosen few*), a genitive (*Hollywood's chosen few*), *some*, *we*, or *us*: the determiners a plural noun takes. Nearly all the lines without one of these have *a*, taken up below.[16]

[15] COCA: *Foreign Affairs*; iWeb: sspnet.org. COCA has five clear lines among typos for *same*, adverbial *some*, titles, and garbled text, and iWeb has 10 hits for `the some who`, most with no earlier *some* in view, so they aren't quotations of the word.

[16] COCA, searched by hand for *lucky*, *chosen*, *fortunate*, *remaining*, and *priveledged* (misspelled in the search) before *few*: of 321 lines with no following noun, 192 have *the*, 90 *a* alone, 25 a demonstrative, 6 a genitive, 2 *some*, 2 *we* or *us*, one a name, and three nothing (a headline, a fragment, and a web post). Web, Indian, and Nigerian sources have more bare cases (*work with* [*select few*] *lenses*; *the group of* [*vicious few*] *who …*, legit.ng), with ordinary adjectives as well as *select*, so the gap may not hold beyond standard English. What differs there may be the article rather than *few*, which I haven't tested; on both analyses these NPs lack a determiner.

Adjectives heading a nominal vary in the same way, word by word. The *rich* type takes *the* or a genitive (*the world's poor*, *our elderly*), though *CGEL* says these NPs are "determined by the definite article" (p. 417).[17] Some of its words take other determiners as well. *Unemployed*, which *CGEL* lists in the type, takes *many* and *some* in edited news ([*many unemployed*] *say they are having trouble getting hired*, *Washington Post*, 2021, NOW), while *these rich* and *some educated* turn up mostly in blogs and in news outside Britain and the United States.[18] *Blue* can head a nominal with no determiner (*I like blue*), and *second* one after *a* (pp. 416–417). Both analyses state the range word by word.

What ordinary Head gives is the same function for *few* in each of these uses: *few would dare*, *few of them*, *the few*, and *the lucky few*. In each, *few* heads the nominal and has no other function; the determiner, if there is one, is another word, as *the* is in *the few*. Fusion gives the first two Det–Head and the last two Mod–Head, so the same word heading the same kind of NP gets a different pair of functions depending on whether a determiner precedes it.

Which adjectives can head a nominal is stated the same way on both analyses. *CGEL* gives types, such as superlatives (*the tallest*), ordinals (*the second*), colours (*I like* [*blue*]), and the special interpretation of *the rich* (pp. 416–417), and excludes **the advancing* (p. 415), **a bigger*, and **the main* (p. 416). *Main* is attributive-only (p. 558), so modifier is its only function, and its exclusion is *CGEL*'s evidence that "the possibility of fusion with the head must in the last resort be stated in syntactic terms" (p. 416). Ordinary Head states the same, as permissions to head a nominal.

The genitive gives no evidence for the modifier function. *CGEL*'s argument from the genitive concerns the determiner function (p. 420), and it states no case requirement for modifiers, so the check here is this paper's own: whether the genitive divides the fused modifier–heads, some taking it and others not, as it divides the determiner–heads (Section 3). It does. *The latter's* and *the former's* are well attested, *the poor's* occurs in edited prose ([*the poor's*] *material well-being*, *Newsweek*), *the rich's* only on the web, and *few* after a determiner almost never takes it.[19] Here too, as with the determiner–heads, the functions these heads share don't predict which take the genitive.

Mod–Head is also where the advantage Section 3 conceded to fusion has to be weighed. A bare determinative takes no pre-head adjective outside a few exclamations, taken up below (**Lucky few escaped*),[20] and fusion derives the gap from adjacency: with *few* as Det–Head, *lucky* would be in the nominal and yet before the determiner. After *the*, *few* is Mod–Head and *lucky* modifies the nominal it heads. Ordinary Head states a condition: a nominal headed by a determinative takes a pre-head modifier only when the NP has a determiner (*hardly* in *hardly anyone* modifies the determinative, inside its DP).

[17] In 500 COCA lines of `[apps*] poor|rich|elderly`, 19 have the adjective heading the NP after a genitive pronoun (*care for* [*their elderly*], *New York Times*).

[18] NOW has 34 lines of *unemployed* heading the nominal after *many* or *some* (`many/some unemployed [v*]`); `these rich [v*]` gave 14 across COCA, iWeb, and NOW, and *some educated* three in NOW. Glass (2019, p. 7) gives further web examples (*Some fired say …*, *Many educated are far from role models*).

[19] In one capped COCA display of *the latter*, *the former*, *the other*, *the poor*, and *the rich* followed by *'s*, *the poor* has 8 genitives (five in magazines and academic writing) and *the rich* 3 (all web or blog). `the few/many 's` gave none, and `few 's` one after a determiner ([*the first few's*] *salty freshness*, *Ploughshares*). A cardinal after a determiner takes it freely: *the two's* in 21 of 34 lines, ten in edited prose.

[20] *Precious few would dare* isn't an exception. For *precious* in *They had precious little money*, Matthews (2014, p. 102) offers the predicative use as the crucial evidence that it's an adverb (*The money they had left was precious little*), and the same test works for *few* (*Volunteers were precious few*). So *precious* is a modifier in the DP, as *very* is in *very few*, not in the nominal.

A lucky few looks like an exception to both fusion's derivation and ordinary Head's condition. Of the 321 COCA lines with an adjective before *few* and no noun after it, 90 have *a* as the only word before the adjective, and where a verb agrees with the phrase, it's plural: [*a lucky few*] *were sent far and wide …* (*Washington Post*). The adjective can have its own modifier (*A very lucky few*, CNN). *CGEL* has two analyses that could apply. In one, *a few* is a complex determinative whose internal structure is "sui generis", admitting *very*, *good*, and *fair*, and it's put among the lexical determinatives "as with the cardinal numerals" (p. 392). In the other, *a* begins a sequence such as *a further* or *a good*, which allows a plural head only if a numeral or *few* quantifies the head, as in [*A further few volunteers*] *were needed* (p. 353), and in *a marvellous hundred days* "*a* is determiner and quite independent of *hundred*" (p. 385 n. 26).

Neither of *CGEL*'s treatments of *a few* favours fusion or ordinary Head. If *a* is the NP's determiner, *a lucky few* is built like *the lucky few*: *lucky* modifies the nominal after the determiner, so adjacency is undisturbed and ordinary Head's condition is met. If *a* belongs to the determinative, so does *lucky*: it modifies *few* inside the DP, not the nominal. Either treatment needs a statement, a list of the sequences with *a* that take a plural head or a slot after *a* inside the determinative, and fusion and ordinary Head need the same one. That's all the comparison needs: on either treatment, *a lucky few* leaves the gap before a bare determinative standing.

For PHP (p. 584), a condition like ordinary Head's is "precisely the kind of stipulation which is not required when the determiner and head functions are fused". They say it of a silent-head analysis of the compounds, but it applies to ordinary Head's condition too. Whether the grammar needs such a statement anyway turns on the other heads with no determiner in their primary use, pronouns and proper nouns, and on three questions: whether such a head takes a pre-head dependent at all, whether it takes a non-restrictive adjective, and whether a restrictive one needs a determiner.

On the first, the grammar already makes a statement of this kind. Pronouns, which by definition "do not permit determiners" (p. 429), "do not normally allow internal pre-head dependents" (**Extravagant he bought a new car*, p. 430), though they take some after the head (*you who worked on both the projects*, p. 429). The compounds pattern the same way, with dependents after the head but not before, and for them *CGEL* derives the gap from fusion (p. 423); a pronoun has no determiner function to derive it from. Ordinary Head's condition differs from the pronouns' restriction only in mentioning a determiner, which pronouns never take. On this question, then, ordinary Head's statement looks like one the grammar already has, and fusion's derivation saves only that one.

On the second, proper nouns take a few non-restrictive adjectives with no determiner (*poor Henry*; p. 520), and the core personal pronouns take *lucky*, *poor*, and *silly* (*Lucky you!*; p. 430). In this exclamation a singular count noun doesn't need its determiner either (*Poor man!*, *Lucky girl!*), and some fused determiner–heads take the adjective too: the compounds (*Poor everybody!*, *Poor somebody!*) and *all* and *both* in partitives. Other partitives are untested. Five are attested, four just after the same exclamation with a proper noun or a pronoun: *Poor Val. Poor everybody.* (*New Yorker*), *Lucky you. Lucky everyone.* (*Everwood*),[21] *Lucky you. … lucky both of us actually.* (*Interior Collective*), *"Poor Ev," the father concludes. … Poor all of us.* (*OC Weekly*), and, answering a mention of *somebody's character*, *"Poor somebody!"* (F. Anstey, *The Giant's Robe*, 1884). Simple determinatives standing alone don't

[21] COCA, `poor|lucky|silly everybody|everyone|somebody|someone`: these 2 of 28 lines; in the rest the adjective is predicative, comes before a clause (*You're lucky someone called it in*), or follows a determiner, and no line has *somebody* or *someone* in the use.

take it, definite or not, even there (**Lucky few!*, **Poor many!*, **Lucky you. Lucky both!*, **Poor Ev. Poor all!*), so the gap doesn't come from an indefinite referent.[22]

For *CGEL* the compounds are fused determiner–heads, for which a pre-head dependent is "not possible" (p. 423), and so are *all* and *both* in partitives (pp. 376, 412), so adjacency excludes the adjective there too. Fusion could make the compounds nouns by conversion, like *a nobody* (p. 423 n. 43), though that won't extend to *all of us*, or list both as exceptions to adjacency. If it takes the adjective as a modifier of the whole NP, though *CGEL* treats *lucky* in *Lucky you!* as a pre-head dependent (p. 430), ordinary Head can do the same, and the exclamation says nothing about adjacency. Otherwise ordinary Head adds the compounds and these partitives to *CGEL*'s exception for "a few adjectives such as *lucky*, *poor*, *silly* with the core personal pronouns", and keeps simple determinatives standing alone out of it. Each analysis needs a further statement, and as analysed here the exclamation gives neither a clear advantage, though the evidence is five attestations and my judgments.

The third question is where pronouns and proper nouns look most like determinatives, and the likeness comes from definiteness. A restrictive adjective after a determiner is routine with personal pronouns (COCA has 350 tokens of *the real me*). *CGEL* would presumably call the word a common noun here, as it calls *he* and *she* in *Is it a he or a she?* (p. 429 n. 46), though the reflexive suggests a secondary use of the pronoun (*The 12-year-old me is kicking myself*). Countability isn't why it needs a determiner, since plural *us* needs one too (*the real us*, **Real us never showed up*). For proper nouns *CGEL* gives the reason: in their primary use names are "inherently definite" and so don't select determiners as ordinary heads do, while in secondary uses that definiteness is lost and determiners are "selected in the ordinary way" (p. 520), as in *the young Isaac Newton*, with *young* restrictive (p. 521). A pronoun restricted to one manifestation of its referent needs a determiner in the same way.

A plural common noun needs no determiner, since its bare form has a reading of its own, and a name's secondary use can be such a noun (*Lucky survivors were airlifted out*, *Young Kennedys still go into politics*). *Few* can't: its use for a group, as in *the few* and *a few*, never occurs bare (*Few were the hangers-on* can't mean that the small group were). *Lucky* modifies that use, so it needs a determiner (**Remaining few were the hangers-on*, beside *The remaining few were the hangers-on*, *Christian Century*). Fusion's adjacency predicts the gap only if bare *few* is a determiner–head, not a modifier–head (Section 3), and the same fact rules out the modifier–head. Given it, fusion gets the gap from structure and ordinary Head from what the adjective modifies. The compounds go the same way: with an adjective they're nouns for *CGEL* (p. 423 n. 43), count with a determiner (*a new everything*, *Rolling Stone*) or non-count without one (*instant everything*, *New York Times*), while as determinatives they take none (**nice anyone*).

So ordinary Head's condition is of a kind the grammar already has, with a source in what the adjective modifies, and fusion's derivation of the pre-head gap, the one fact of form that favours the determiner function (Section 3), saves less than PHP suggest.

Nothing in the form of Mod–Head expressions, then, shows the modifier function.

[22] This assumes the exclamation has no silent determiner, which would put *few* after a determiner, where both analyses allow the adjective (*the lucky few*). *CGEL* gives the comparable NPs none: pronouns "do not permit determiners" (p. 429), *lucky* in *Lucky you!* is a pre-head dependent (p. 430), and vocatives take a bare singular count noun with an adjective (*old chap*, *young man*; pp. 522–523).

5 *Once*, *twice*, AND *thrice*

PHP built their case for fusion on *once*, *twice*, and *thrice*. They distinguish five uses of *once*: numerical (*I will only say this once*), singulative ('a single time': *Not once have I seen anything like this*), past (*I was once a teenage werewolf*), term of office (*the once mayor*), and sequential (*once he arrived*), which they analyse as a preposition like *after*, with no fusion (pp. 584, 592, 594). *Twice* and *thrice* have the numerical and term-of-office uses, and a multiplier use (pp. 594–596).

Much of their argument concerns category. Against an adverb analysis, numerical *once* takes determiners and relative clauses (*for every once that the large pulley goes round*, p. 586 and n. 28) and can function as subject (*Once is enough*, p. 587); against a noun analysis, it takes the pre-head modifiers of *one* (*at least once*, *more than once*, p. 589). It's a determinative, and on ordinary Head it heads the nominal of its NP, so only the second function is in question.

They take the form of the word to show that function too. In *someone* and the other compound determinatives "the morphological compounding in this case reflects the fusion of two syntactic functions", and *once* is "similarly" a compound, of a determinative base *on·* and a noun base *·ce* 'time' (pp. 587–588). But simple *some* and *few* are fused with no compounding, and PHP's own note takes *·ce* to be a reanalysed genitive ending (p. 588 n. 30), and Wu (2021, p. 89) finds their noun base "far from convincing". The two bases can as well reflect the word's meaning, a quantifier with a sort, as in the compounds (Section 3), and both analyses give *once* that meaning.

The first thing fusion is said to explain is why *once* occurs both with and without a determiner, which a singular count noun doesn't. Numerical *one* functions as determiner (*one time*) and as modifier after a determiner (*this one time*), so *once* is a fused determiner–head on its own and a fused modifier–head after *this* (p. 588). That's the link between the two uses that Section 4 examined for *few*, a default with exceptions both ways. Ordinary Head states it as a permission: *once* may head a nominal with or without a determiner.

The second is the adjective. Since the fused *once* is also the determiner, **last once* is excluded "for the same reason as" **last one time*, while *this last once* is fine (p. 589). One of their *JL* referees judged *this last once* ungrammatical, and they allow that speakers may vary (p. 586 n. 26). The searches of COCA, NOW, and iWeb found no relevant example of *this last once* or *the last once*, but they found the pattern with *twice*, their (33b) (p. 595), as in [*The last twice*] *I checked my 71 GT*... (mgexp.com).[23] So the adjective after a determiner is attested, though not with *once*.

The gap before a bare *once* is the one Section 4 weighed for *few*.[24] Fusion derives it from adjacency, and ordinary Head excludes it by its condition on pre-head adjectives (Section 4). Their further example, *(*this*) *once last* with a silent head after *last* (p. 589), separates fusion from a silent head, not from ordinary Head.

Larger phrases and the other uses go the same way. A DP such as *more than once*, or a PP such as *up to twice*, is for PHP a fused determiner–head on its own and a fused modifier–head after a determiner (pp. 589–591 and n. 31), and ordinary Head puts each in the head function alone. Singulative and past *once* take no determiner and no attributive adjective (p. 592), and both gaps fall under what's

[23] The hits for *once* are the verb *last*, slips for *one* or *ounce*, and adverbial or sequential *once*. NOW has 290 tokens of *the last twice*, 176 of the 180 displayed from British or Irish sources, mostly racing reports (*He's gone well* [*the last twice*], *Racing Post*); iWeb has 28 such lines of 39, COCA none.

[24] None of 97 COCA lines of *a*, a word tagged as an adjective, and *once* or *twice* has the pattern of *a lucky few*, so nothing here calls for the statement *a lucky few* needs on both analyses.

already been weighed: the missing determiner as a lexical restriction stated on both analyses, the adjective under the trade counted for *few*.

Term-of-office *once* functions as attributive modifier (*the once mayor*), as does *twice* in *the twice president*, which many speakers find unnatural (pp. 593, 595). Both analyses have to state that the use is largely confined to nouns for offices (**my once job as a glass-blower*, pp. 585, 593). For PHP, *once* here "will act as a fused modifier-head forming a nominal rather than forming a full NP" (p. 593), so the attribute is a nominal whose head is also its modifier. On ordinary Head the determinative *once* simply functions as modifier in the nominal *once mayor*, as *two* does in *the two books*. That's one function where fusion has two, and one nominal fewer, though the lexical statement is no shorter.

The rest of their case (multiplier *twice*, *once a week*, and the non-standard uses, pp. 595–597) turns on category and on the word heading a nominal, which the analyses share.

Apart from the adjective restriction weighed in Section 4, no property of form in this paradigm varies with the second function rather than with the word, so on the criterion of Section 2 the paradigm fusion was developed on adds no evidence of its own.[25] Where their structures differ, in the term-of-office use, ordinary Head has the fewer functions.

6 What each analysis states

Sections 3–5 compared the two analyses construction by construction. Table 1 puts the results together. Both columns use *CGEL*'s categories and the same lexical facts, so what the table compares is how each analysis has to state them.

Below the line is the trade, and fusion has a good case. The grammar keeps constituents with two functions for fused relatives anyway (Section 7), so giving one to a determinative that stands alone adds no new kind of structure, and it derives a restriction ordinary Head has to state: no pre-head modifier in a nominal headed by a bare determinative. The added function matches the word's dependent use (*few* is determiner in *few students* and modifier in *the few students*), which keeps fusion's licences general, but the match is how the function is chosen, not evidence for it. Ordinary Head states the condition instead, and for *few* and the compounds it follows from the use the adjective modifies, as with a restrictively modified pronoun or name (Section 4). What ordinary Head gains is parsimony: *few* has one function in all its uses as head, and constituents with two parents stay in one construction. The compounds' restrictor, which PHP also credit to fusion, adds nothing to either side (Section 3).

For the determiner function the comparison available is between a determinative standing alone and the same word after a determiner (*few* against *the few*), since *CGEL* gives determinatives no use as a pure head, and it changes two things at once: the word loses its determiner function, and a determiner precedes it. A fact that separates the two uses could follow from either change, so it counts for the determiner function, but for less than a contrast in the function alone would.

The pre-head adjective does show something, and it counts for fusion with Det–Head, the analysis it also favours over Mod–Head (Section 3). But it's one fact, and it weighs less than it seems, since both analyses need *few* to lack a bare use for a group (Section 4). The exclamation doesn't add a second fact: there the compounds and partitive *all* and *both* take the adjective and simple determ-

[25] COCA, NOW, and iWeb have no genitive of *once*, *twice*, or *thrice*. Ordinary Head adds them to its list of gaps. Fusion predicts the gap for the Det–Head uses, unless a compound's nominal base can carry the genitive, as in *everyone's* (Section 3), and then it would have to set *-ce* apart from *-one*.

Table 1: What fusion and ordinary Head state for the heads *CGEL* fuses: determinatives, adjectives, nouns such as *nylon*, and *once*, *twice*, and *thrice*. Above the line, the two analyses state the same facts, each in its own terms; below it, they differ.

	Fusion (*CGEL*)	Ordinary Head
Determinatives that stand alone (*some*, normally not *every*)	A list of the words that allow Det–Head	The same list, as permissions to head a nominal with no determiner
Determinatives that head a nominal after a determiner (*the few*)	A default from the modifier use, with exceptions both ways: head but not modifier (*some*, *both*), modifier but not head (*every*, *various*)	The same default and exceptions, as permissions
A restrictor (*something reliable*, *little new*)	Derived from fusion (p. 423), but kept after conversion, and after *few*, *some*, and *none* limited to a comparative or an adjective such as *present* (Section 3)	Permitted after the compounds, *all*, *much*, and *little* as head, and the same limit after *few*, *some*, and *none*
The NP's determiner (*the few*, *the rich*)	Word by word: a plural noun's range for *few* (*those chosen few*); *the* (p. 417) or a genitive for *rich*, and also *many* or *some* for *unemployed*; none needed for *blue* (*I like blue*)	The same
A few and its expansions (*a lucky few*)	On *CGEL*'s treatment with *a* as determiner, a list of sequences with *a* that take a plural head (p. 353); on its complex-determinative treatment, a slot inside the determinative (p. 392)	The same
Adjectives that head a nominal	*CGEL*'s types, with exclusions such as **the main* (pp. 415–417)	The same types and exclusions, as permissions
Genitive on a determinative standing alone	A gap predicted for all these words, with a list of exceptions (*either's*, *neither's*, *each's*, *another's*; compounds such as *everyone's*, unless their nominal base carries it)	No general bar, with a list of gaps (*all*, *many*, *few*, *some*; *once*, *twice*, *thrice*)
Change of category on the head path	Where a DP, AdjP, or PP heads a nominal (*up to twice*, PHP, p. 589 n. 31)	The same, except term-of-office *once*
Once, *twice*, and *thrice*	Numerical uses Det–Head alone and Mod–Head after a determiner; singulative and past uses Det–Head only (PHP, pp. 588, 592); sequential use a preposition (p. 594)	The same uses, as permissions to head a nominal with or without a determiner (numerical) or only without one (singulative, past); sequential use a preposition
A second function for the head of the nominal	Determiner for a determinative with no determiner before it; modifier for one after a determiner, and for adjectives and nouns such as *nylon*; term-of-office *once* a nominal of its own as attribute (PHP, p. 593)	None outside fused relatives (Section 7); term-of-office *once* a modifier, as *two* is in *the two books*
A pre-head adjective with no determiner (**Lucky few escaped*)	Excluded by adjacency (Section 3; p. 423), with an exception for the compounds and partitive *all* and *both* in an exclamation (*Poor everybody!*, *Poor all of us!*; Section 4)	Excluded by a condition, only after a determiner, which for *few* and the compounds follows from the use the adjective modifies, as with pronouns and names (Section 4); in an exclamation the compounds and partitive *all* and *both* join the pronouns' exception and simple determinatives standing alone don't (**Lucky few!*)

inatives standing alone don't, a line neither analysis predicts. Fusion's structural route and ordinary Head's fewer functions are economies of the grammar, not facts of form, and the comparison is close. The burden Section 2 puts on the second function decides it: I take the balance to favour ordinary Head, as a reasonable preference, not a forced conclusion.

A reader who weights derivation above parsimony will take fusion for the simple determinatives, and nothing found here rules that out. The balance would shift if adjacency predicted further restrictions on bare determinatives, beyond pre-head modifiers in the nominal, that ordinary Head could only list.

The saving is in where fusion applies, not in whether the grammar has it: the grammar these results favour keeps constituents with two functions for fused relatives and removes them from noun phrases without a relative clause, giving each word fewer functions in its uses as head.

7 Fused relatives: where the second function shows

Sections 3 and 4 found one fact of form that favours a second function in Det–Head expressions, the missing adjective, and none in Mod–Head ones. Fused relatives are the remaining test. *CGEL* brings fusion to them in so many words: "we invoke the same concept of functional fusion in our analysis of fused relatives" (p. 1073).[26]

(4) a. *I spent* [*whatever he gave me*]. (*CGEL*, p. 1068)
b. [*Whoever told you that*] *can't have read the report.* (p. 1076)
c. *I'll go* [*wherever you go*]. (p. 1068)

In (4a), *whatever* is head of the object NP and prenucleus (the fronted position; Section 2) of the relative clause, with a gap after *gave me* (*CGEL*, p. 1073; Figure 3a). In (4b), *whoever* is head of the subject NP and also the subject of *told you that*, which PHP (p. 579 n. 16) call "fused head-subject". The criterion asks whether the function inside the relative clause shows in form.

Ordinary Head is available here too. *Whomever I see* could be an NP headed by *whomever*, with a bare relative as modifier in its nominal, just as *person* heads *the person I see*. Put in *CGEL*'s terms, it's the analysis Bresnan and Grimshaw (1978) proposed: the relativized phrase heads the whole NP, and "there is no fronted element in the relative clause" (PHP, p. 578). Bresnan and Grimshaw also require the head to agree with the gap, a condition taken up below with case. At first sight the analysis needs nothing beyond ordinary Head, a kind of relative clause the grammar already has, and that condition, and I'll call it the head analysis (Figure 3b).

[26] Transparent free relatives, as in *He made* [*what may appear to be a radically new proposal*] (Grosu, 2003, p. 248), raise different questions and are set aside here.

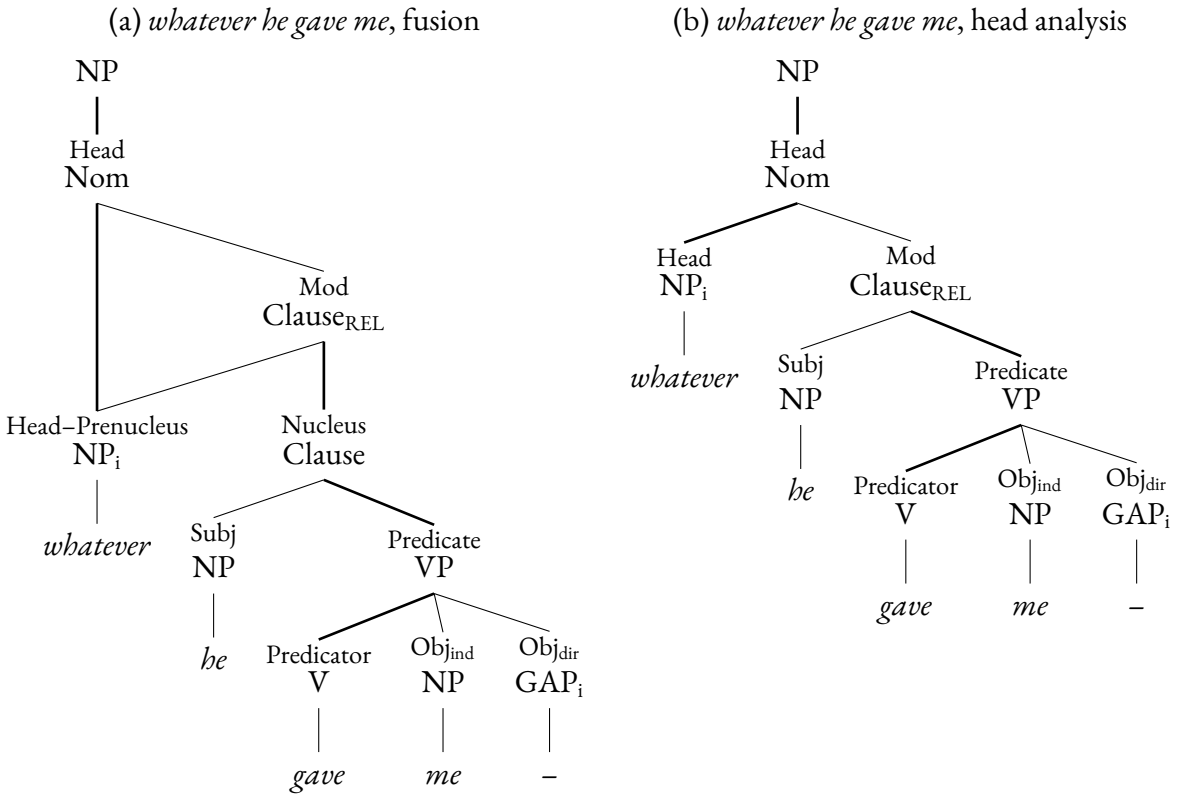


Figure 3: A fused relative under fusion (a), following *CGEL*'s tree for *what she wrote* (p. 1073), and under the head analysis (b), adapted from Bresnan and Grimshaw (1978). In (a) the NP *whatever* has two parents, the nominal it heads and the relative clause in which it's the prenucleus, linked to the gap. In (b) the NP heads the nominal, outside the bare relative clause that is its modifier. Thick lines join heads to their parents, the nucleus counting as head of the relative clause. Other NPs are shown without their structure, as in *CGEL*'s tree.

Much comes out the same. The evidence that fused relatives are NPs holds on either: agreement, subject–auxiliary inversion, no extraposition, a preposition left stranded (*what she referred to*, not **to what she referred*), and the functions of NPs (*CGEL*, pp. 1068–1070). So does what the literature calls matching, in which the relativized phrase's category suits "categorial selection both in the matrix clause and in the relative clause" (van Riemsdijk, 2006, p. 349). Bresnan and Grimshaw (1978, pp. 336–338) derive the match with the matrix from headedness, since a phrase has the category of its head, and on either analysis the match with the gap comes from the relative clause. So category matching doesn't separate the two analyses, and nor do the *-ever* forms PHP cite as a formal mark of fused relatives (p. 576), which go with the construction both analyses recognize. Three other properties do, the third less directly.

First, in standard English bare relatives can't relativize their subject. *CGEL* contrasts [*The car I took*] *was Ed's* with *[*The car hit us*] *was Ed's*, and the restriction holds whatever the head: ![*Anyone wants this*] *can have it* is non-standard (p. 1055).[27] *Anyone* is a compound determinative and the

[27] *CGEL* gives two less certain cases, in an *it*-cleft and an existential (?*It was my father* [*did most of the talking*], ?*There's someone at the door* [*wants to talk to you*]), which fall "at the boundary between very informal and non-standard" (p. 1055).

ultimate head of its NP on either analysis (Section 3). So that example has the head analysis's structure, a head with a bare relative, and it fails. Holding the frame constant gives the same contrast with a fused relative:

(5) a. *I will do business with* [*whichever of them comes up with the cash*]. (PHP, p. 579 n. 16)
b. !*I will do business with* [*any of them comes up with the cash*].
c. *I will do business with* [*whichever of them you recommend*].
d. *I will do business with* [*any of them you recommend*].

With an object gap, *any of them* takes a bare relative freely (5d), so what rules out (5b) is subject relativization. On the fused analysis, *whichever of them* is itself the subject of *comes up with the cash*.[28] The clause has no subject gap, and the restriction doesn't apply. The head analysis has to give relative words a permission that the restriction withholds from every other head.

The second function needn't belong to the relative word alone. The contrast is the same with a noun, as in *cutting the sponge up into* [*whatever size fits*] (*Coral Sea Dreaming: Awaken*) against !*cutting the sponge up into* [*any size fits*]. There *whatever* is only the determiner, so the second function belongs to the NP *whatever size*, whose ultimate head is the noun, as PHP (p. 572 n. 8) say of *whatever meals*.

Second, other heads take infinitival relatives without a relative word, as in *the ideal person* (*for you*) *to confide in* (*CGEL*, p. 1067), but a fused relative can't be infinitival: **I can't afford* [*what to buy*] (p. 1073). Kim (2017, p. 364) also gives **He is still wearing* [*what to wear to bed*].

Without a further restriction, the head analysis allows *what to buy*, as it allows *a thing to buy*. On the fused analysis, *what* would be the relative phrase of an infinitival relative clause, and there "the relative phrase must consist of preposition + NP" (p. 1067), as in *the ideal person* [*in whom*] *to confide*. The fused element of an NP fused relative is an NP, so it never qualifies.[29]

Third, case. *CGEL* names case as the form in which a second function would show (pp. 420, 1073–1074), and for fused relatives with *who* and *whoever* it states the requirement: the pronoun "has to satisfy the case requirements of both the relative clause and the matrix clause" (p. 1073).

(6) a. [*Whoever is responsible for the damage*] *must pay for it.* (*CGEL*, pp. 1073–1074)
b. *He will criticise* [*whomever she brings home*].
c. ?[*Whomever he marries*] *will have to be very tolerant.*

It ties them to position: there the relative clause "could not be misconstrued as predicate of the matrix clause". A fused relative as subject is where that misconstrual could arise ([*Whoever told you that*] *can't have read the report*), so these cases give the head analysis no parallel.

[28] On *CGEL*'s analysis, *whichever* is fused twice here. Inside its own NP it's a fused determiner–head (*CGEL*, p. 398), and that NP is fused head and subject in the relative construction, which PHP count as "multiple instances of fused functions" (p. 579 n. 16). *Whichever* alone, in *I'll take* [*whichever he gave me*], is fused twice in the same way, with head and prenucleus. Ordinary Head removes the inner fusion, for which the only evidence of form is the missing pre-head adjective (Section 3), and keeps the outer one, which this section's three properties show (Section 6; Figure 4).

[29] *CGEL* finds "no evident explanation" for that restriction (p. 1067), and the argument takes it as stated. *CGEL* also says infinitival relatives with a relative phrase occur only as integrated relatives (p. 1067), which would exclude fused ones directly. The restriction on the relative phrase covers NP fused relatives without that statement. It doesn't cover a PP fused relative whose relative phrase is a preposition with an NP, as in *I'll work* [*in whichever city they want me to work*] (PHP, p. 575). That **I'll work* [*in whichever city to work*] is out too rests on the statement about integrated relatives.

d. ?*She lunches with* [*whomever is going her way after morning classes*].

In (6c), the relative clause calls for accusative and the matrix for nominative. In (6d) it's the reverse. *CGEL* judges both "at best very questionable" (p. 1074).

On the head analysis as stated so far, *whomever* stands outside the relative clause, so its form should answer to the matrix alone. That was the analysis's attraction for matching: the head "would automatically bear the case of the whole NP" (PHP, p. 578). Bresnan and Grimshaw (1978)'s agreement condition between the head and the gap is where case could come in, and they report that languages differ on whether case is among the features that must agree (pp. 353, 373–374). With case included, the head analysis can make *whomever* answer to the clause too, but only by a further statement, where fusion gets the same result from the relative word's function inside the clause. Without case in the condition, the head analysis predicts (6c) to be degraded but not (6d). There *whomever* is the object of *with*, where accusative is the formal choice (*CGEL*, pp. 465–466). The question mark on (6d) shows the subject function inside the relative clause.

Usage shows the same thing, as a preference. The figures show which form speakers and writers choose, not which is acceptable, so this evidence is graded where the first two properties are categorical. After a preposition the matrix function, the function of the whole fused relative, is the same in nearly every token, so the head analysis without the agreement condition gives no reason for the choice of form to differ with what follows. In COCA, *whoever* and *whomever* occur after a preposition 802 times before a pronoun subject, so with a non-subject gap, as in *with whomever they …*, and *whomever* has 395 of those (49%, or about 52% once the gaps are checked). They occur there 2,458 times before a verb, where the gap is the subject, as in *to whomever is …*, and *whomever* has 352 of those (14%).[30] Both counts follow a preposition, so attraction to the preposition is the same in each, and the difference between 49% and 14% follows the gap inside the clause.

So the function inside the relative clause shows in the form. The matrix function shows too, and not only after prepositions (Table 2). As the subject of the main clause, *whomever* is rare even with a non-subject gap ([*Whomever I chose for this experiment*] *should be close to the mean*, *Southern Review*).[31]

[30] COCA, `[i*] who*ever [p*]` and `[i*] who*ever [v*]`. Tokens before *it* are left out, because *it* can be the subject of a clause whose gap is a predicative complement, as in *whoever it is*. The verb list shows only its 1,000 most frequent types, so its counts are lower bounds. A check of about 200 lines of each form found a non-subject gap in all but one *whomever* line before a pronoun other than *it*, and in 131 of the 148 such *whoever* lines, the rest having a predicative gap (*for whoever she is*), the subject of an embedded clause, or a fused relative that's the subject of a clause after *until* or *before*. Without them the share is about 52% (about 393 of 753); further checks are in the evidence index.

[31] In COCA, for the main clause's subject, all the *whomever* hits after a full stop and 500-line samples of the *whoever* hits, leaving out exhaustive conditionals (*Whoever you are, …*), interrogative for *CGEL* (p. 1072 n. 15); for a verb's object, about 200 lines of each form after a lexical verb, leaving out predicative gaps (*find whoever it is*) and counting content-clause subjects apart. The queries and counts are in the evidence index. The 8-of-51 cell is small, so its share is rough.

Table 2: The share of *whomever* among tokens of *whoever* and *whomever* in COCA, by the function of the fused relative and the gap in its relative clause, in the search frames used: a preposition, a lexical verb, or the start of a sentence before the form, and after it a pronoun other than *it* (non-subject gap) or a verb (subject gap). Figures marked "about" are estimated from labelled samples of capped displays. The cell for a preposition's object with a subject gap counts frequency-list rows, which are lower bounds, so its share isn't a bound. The 51 in the last row is 8 *whomever* and about 43 *whoever* (33 in a 498-line sample of 644).

Fused relative	Subject gap	Non-subject gap
Object of a preposition	14% (352 of 2,458)	about 52% (393 of 753)
Object of a verb	about 6% (60 of about 970)	about 37% (206 of 562)
Subject of the main clause	under 2% (32 of about 2,100)	about 16% (8 of about 51)

In both columns the share falls from preposition to verb to subject, following the function of the whole phrase, and in each row it's several times higher with a non-subject gap, following the function inside the clause. Prepositions and verbs both call for accusative, so the step between those rows isn't a difference in the case the matrix function calls for; in writing alone the non-subject-gap column narrows to 65% against 58%, so part of it is register. Where the cells can be split, the order holds in speech and scripts and in writing taken separately, though *whomever* is far rarer in the first.[32]

So the form answers to both functions. *Whoever* is the default: in the pooled figures, *whomever* comes near half only where both functions call for accusative, and it's a minority in the clashes, one of which is (6c). Fusion with *CGEL*'s case requirement predicts that a clash has no fully acceptable form, which is what *CGEL*'s question marks record; *CGEL* adds that *whoever* "would be preferable in both, but many would regard it as less than fully acceptable in formal style" (p. 1074). Usage shows both functions at work, though how strongly each counts, and how the two interact, I leave open. So case favours fusion here rather than forcing it.

CGEL drew this analogy itself. Its explanation of the missing genitive in **many's expenses* is that the form's "case should match both functions", and it points to "the discussion of case in fused relatives" (p. 420). Among fused determiner–heads, the genitive divides the words: the searches found very few tokens of *many's*, but *either's*, *each's*, and *everyone's* occur. The functions, which all these words share on *CGEL*'s analysis (Section 3), don't predict that division, and *the poor's* puts the genitive on a fused modifier–head (Section 4). In fused relatives the word stays the same and the form follows the function inside the relative clause, as a preference. On the diagnostic *CGEL* itself chose, the two constructions come apart.

CGEL also has PP fused relatives (*I'll go* [*wherever you go*], p. 1068), and PHP (p. 575) add AdjPs and AdvPs (*I'll be* [*however responsive they want me to be*]). The case test doesn't apply to them, but the first property can: in *Come round* [*whenever suits you*], the head analysis would have a bare relative with a subject gap. On the head analysis, that's again a head with a permission no other head has, since "standard relative clauses are modifiers only within Nom" (p. 575).

[32] Where the fused relative is the subject of a verb's content clause (*I think* [*whoever did this to her*] *wants …*, *Wildling*), *whomever* has 9 of about 400 tokens with a subject gap, about 2%, as for the main clause's subject. The cells split into spoken, TV, and film lines against the rest are in the evidence index; fiction has dialogue and web texts are informal, so the split is rough.

So the relative word's place in the clause shows in form in three ways, two of them categorical and one graded. The head analysis can state each, but only by giving fused-relative heads statements no other head needs: subject relativization without a relative word, no infinitival form, and case among the features its head shares with the gap. Fusion gets all three from the rules for relative clauses that have a relative phrase. So fusion is favoured here, though not forced. The subject and infinitival facts show that the relative word has a function inside the clause, but they don't choose among fusion, movement, and multidominance as ways of giving it one.

A dependency grammar draws a related line: Gisborne (2026, p. 2) disprefers mutual dependency where an analysis without it is coherent, and keeps it for free relatives, where the *wh* word heads the construction and is also extracted (p. 16).

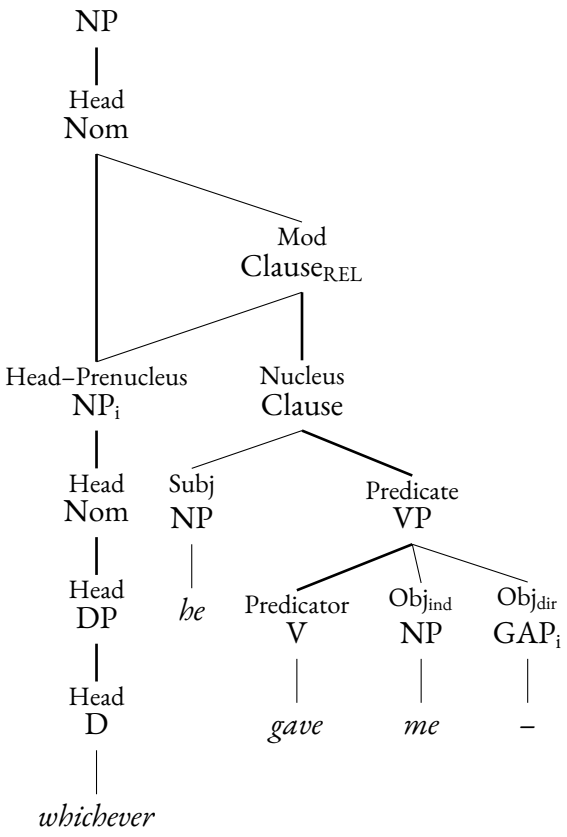


Figure 4: *Whichever he gave me* as this paper analyses it. Inside its NP, *whichever* is a pure head, where *CGEL* has a second fusion (p. 398). The NP is still head of the nominal and prenucleus of the relative clause at once, as in Figure 3a.

8 Conclusion

On the commitments held fixed in Section 2, this paper finds no property of form that shows the modifier function where *CGEL* fuses the head of a noun phrase with a modifier (*the lucky few*, *the rich*), the clearest of its results. PHP's *once*, *twice*, and *thrice* add nothing further (Sections 4 and 5). For the determiner function (*take some*) it finds one, the absence of an adjective before a bare determinative, which fusion derives and ordinary Head states (Section 3). An exclamation that lets a pronoun

take the adjective lets the compounds and partitive *all* and *both* take it too, though not a simple determinative standing alone, so it gives neither analysis a clear advantage (Section 4). Against the gap, ordinary Head assigns fewer functions and keeps constituents with two parents to one construction; the comparison is close, and the balance favours ordinary Head without forcing it (Section 6).

Noun-phrase fused relatives supply the evidence the criterion asks for, in two properties categorically and in a third as a preference (Section 7). The relative word's function inside its clause shows in subject relativization and in the ban on infinitival fused relatives, and in usage the case of *whoever* answers both to that function and to the function of the whole phrase. So fusion survives, favoured though not forced, as a device the grammar licenses for one construction, not as the general answer to a noun phrase with no noun.

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